بسم الله الرحمن الرحیم



استاد : دکتر محمدرضا شایگان مطلق

گردآورنده : فرزان آهنگر

موضوع : ارزیابی کارایی کنترلگر های تعریف شده توسط نرم افزار

Performance Evaluation of Software-Defined Networking Controllers

A Comparative study

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چکیده

SDN چیست ؟

چرا به چارچوبی برای ارزیابی SDN Controller ها نیاز داریم ؟

Evaluating QoS (delay and throughput)

OpenFlow Protocol

Impact of routing protocols in Data Centers

Proper Deployment Of SDN Controllers

مقدمه

Traditional Networks

SDN Decouples

SDN Layers

Data Planes

SDN is Centralized , Vendor-agnostic , Programmable

Northbound API

Southbound API

Best Controller Per Scenario

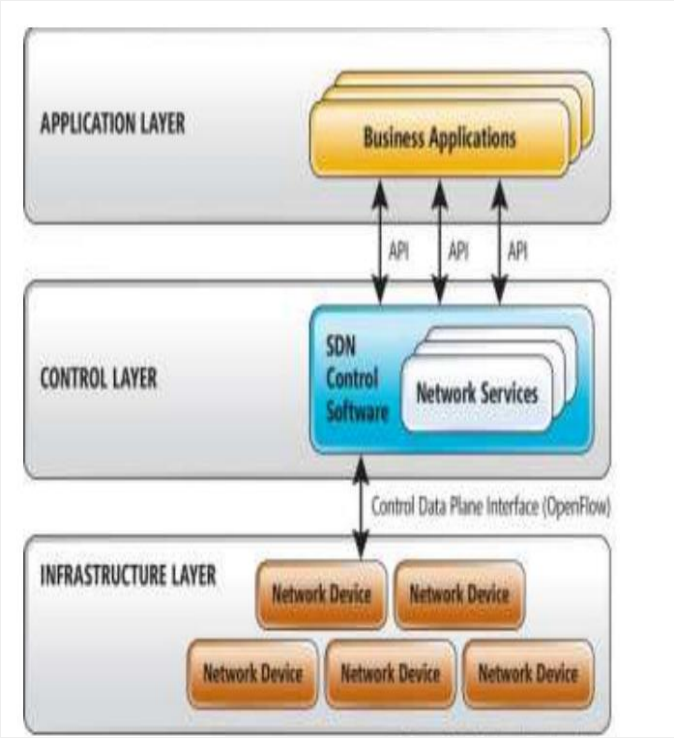
What Controllers We Evaluate

Topologies

Evaluation Types

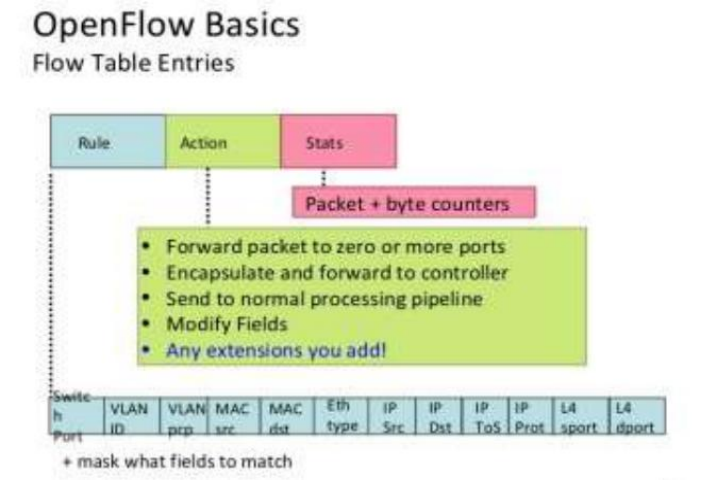
SDN Controllers

- The Control Plane (Controller)
- SB-API
- NB-API
- OpenFlow
- Flow Tables Via SSL/TLS
- Flow With No Match
- Flow Entries



SDN Controllers

- OpenFlow Messages Types
- Controller Types



Version	Features
1.0	Single flow table, L2 + IPv4
1.1	MPLS support
1.2	Flexible matching, IPv6
1.3	Parallel controller-switch channels
1.4	Improved matching, optional ports
1.5	Outgoing table, matching by outgoing ports

Related Work

Evaluators : tootoonchian, Gorbunov, Ganjali, Casado, and Sherwood

- Groups Evaluate Open-Source Controllers → NOX , BEACON , Maestro , NOx & NOX-MT
- Designing a Benchmarking Tool For OpenFlow Switches Called CBench
- Multithreading Of NOX-MT

Related Work

Evaluators : Stancu, Halunga, Vulpe, Suciu, Fratu and Popovici

- Groups Evaluate Open-Source Controllers → RYU , POX , ONOS , OpenDayLight
- Simple Configuration And Environment Suitable Winner
- The Industry Failure And Bad Benchmark
- Lack Of Traffic Models Similar To Data Centers

Related Work

Evaluators : Shiva, Vajihe and Manije

- Groups Evaluate Open-Source Controllers More Precisely And Better → Floodlight , OpenDayLight
- OpenDayLight -> Tree Topology , Low-latency On Half-bandwidth Traffic
- Flood Light -> Tree Topology , Low PL , Heavy Load
- Limited Study Parms , Low Number Of Controllers

Related Work

Evaluators : Darianian, Williamson and Haque

- Groups Evaluate Open-Source Controllers On Both Physical And Virtual Env → ONOS , OpenDayLight
- Better Performance In Trough-put and Delay
- Mostly Used In Data Center And Cloud Env But Not Tested There

Related Work

Evaluators : Fancy and Pushpaltha

- Groups Evaluate Open-Source Controllers Based On Developing Programming Language → POX , Floodlight
- Various Network Topology In Mininet
- But They Found Out Programming Language Is Not Very Important In Case Of Through-put or Delay

Quantitative Comparison of Five Open Flow Controllers

Centralized Controllers

Metrics → QoS , Through-put , Delay

Other Factors → Topology , Scale

Distributed Controllers

Metrics → Network Deley

Other Factors → Topology , Scale

controller	Support partners	NBAPI	SBAPI	Supported platforms	Centralized /distributed	Programming language
ONOS	ON.LAB, AT&T, Ciena, Cisco, Ericsson, Fujitsu, Huawei, Intel, Nec, Nsf.Ntt Communication, SK Telecom	REST API	OF 1.0, 1.3, NETCNF	Linux, MAC OS, and Windows	Distributed	Java
OpenDayLight	Linux Foundation with memberships covering over 40 companies, such as Cisco, IBM, NEC	REST API	OF 1.0, 1.3, 1.4, NETCONF/Yang, OVSDB, PCEP, BGP/LS, LISP, SNMP	Linux, MAC OS, and Windows	Distributed	Java
POX	Nicira	REST API	OF 1.0	Linux, MAC OS, and Windows	Centralized	Python
RYU	Nippo Telegraph And Telephone Corporation	REST For Southbound	OF 1.0, 1.2, 1.3, 1.4, NETCONF, OFCONFIG	Most supported on Linux	Centralized	Pyrhon
Floodlight	Big Switch Networks	REST API	OF 1.0, 1.3	Linux, MAC OS, and Windows	Centralized	Java

Centralized Controllers

RYU , BEACON , FloodLight , POX , NOX

Academic And Research Use

Evaluating RYU , POX Which is an Improved Version

FloodLight Picked

Problems

Distributed Controllers

HYPERFLOW, Elasticon, ONOS, DISCO, OpenDayLight

WAN & Data Centers

Features And Use Cases

Evaluating ONOS , OpenDayLight Because Of

Supported By Well Known Brands Like

Appropriate Evaluation

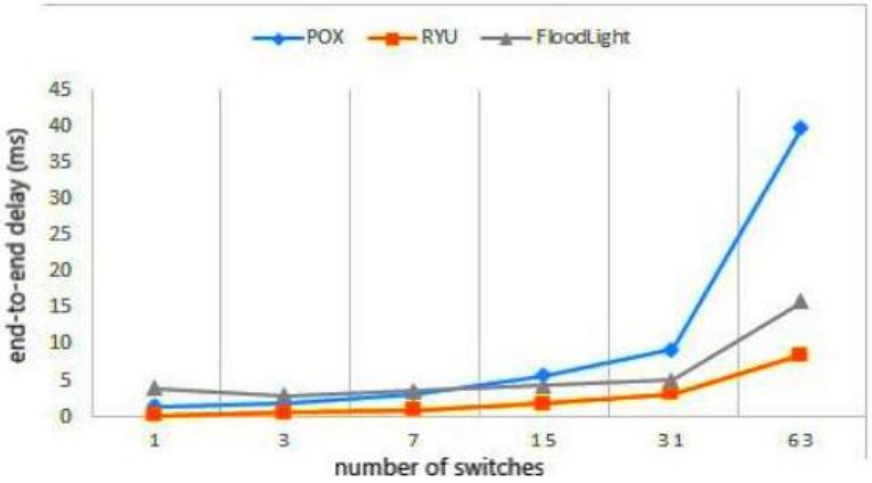
End-To-End Delay

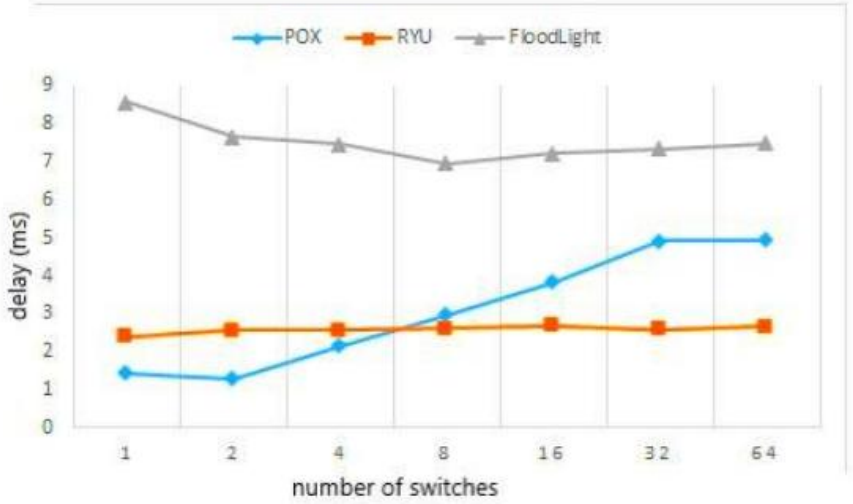
Network Bandwidth

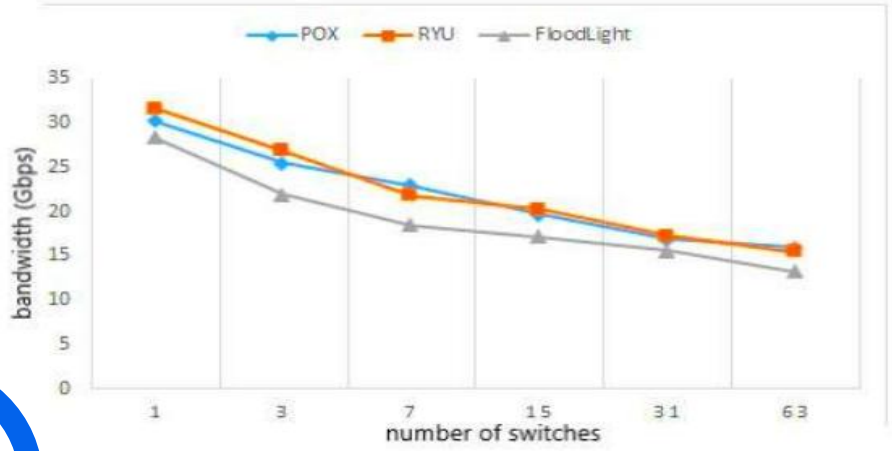
POX , RYU , FloodLight

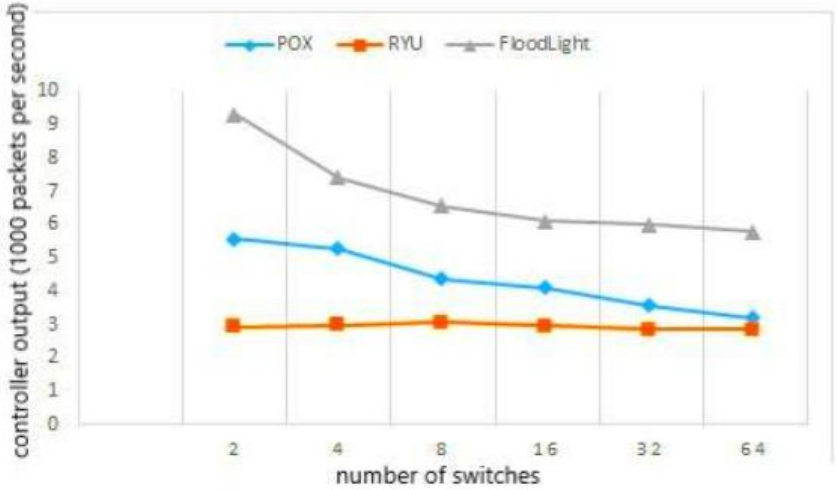
Average of 10 tests
1 to 63 OpenFlow Switches

RYU While Scale Increases Was the Best
POX With Highest Deley


POX
RYU
FloodLight
end-to-end delay (ms)
0
5
10
15
20
25
30
35
40
45
1
3
7
15
31
63
number of switches


POX
RYU
FloodLight
delay (ms)
0
1
2
3
4
5
6
7
8
9
1
2
4
8
16
32
64
number of switches


POX
RYU
FloodLight
bandwidth (Gbps)
0
5
10
15
20
25
30
35
1
3
7
15
31
63
number of switches


POX
RYU
FloodLight
controller output (1000 packets per second)
0
1
2
3
4
5
6
7
8
9
10
2
4
8
16
32
64
number of switches

FlowTable Mechanism

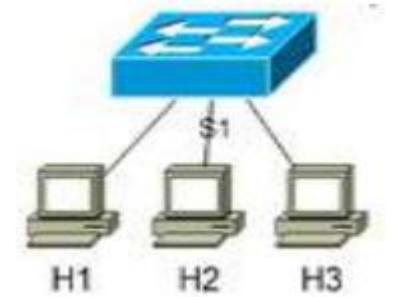
Packets-in Packets-Out

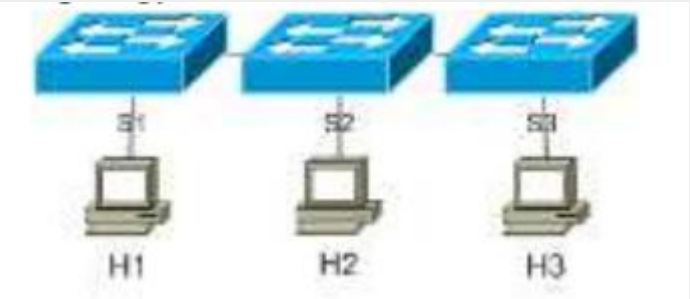
Trough-put

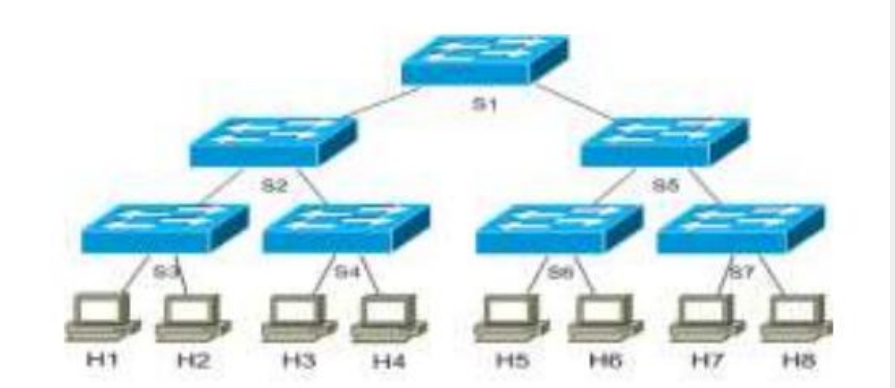
Programming Language Delay

Ryu Static Delay

Effective Network Variables on Performance



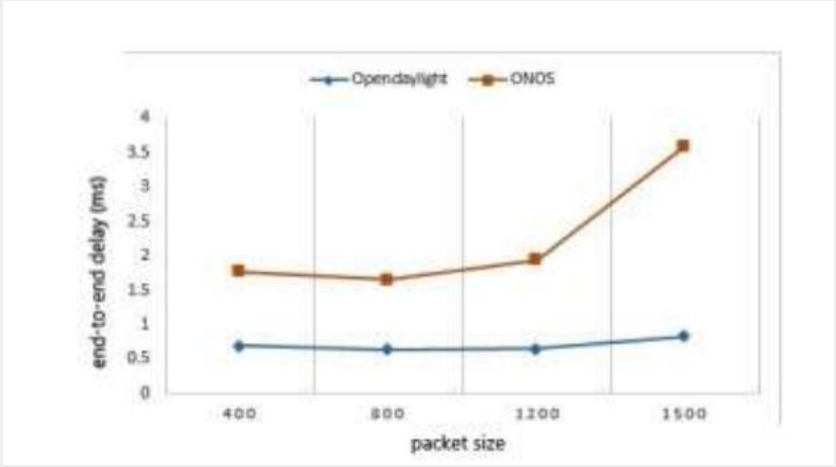


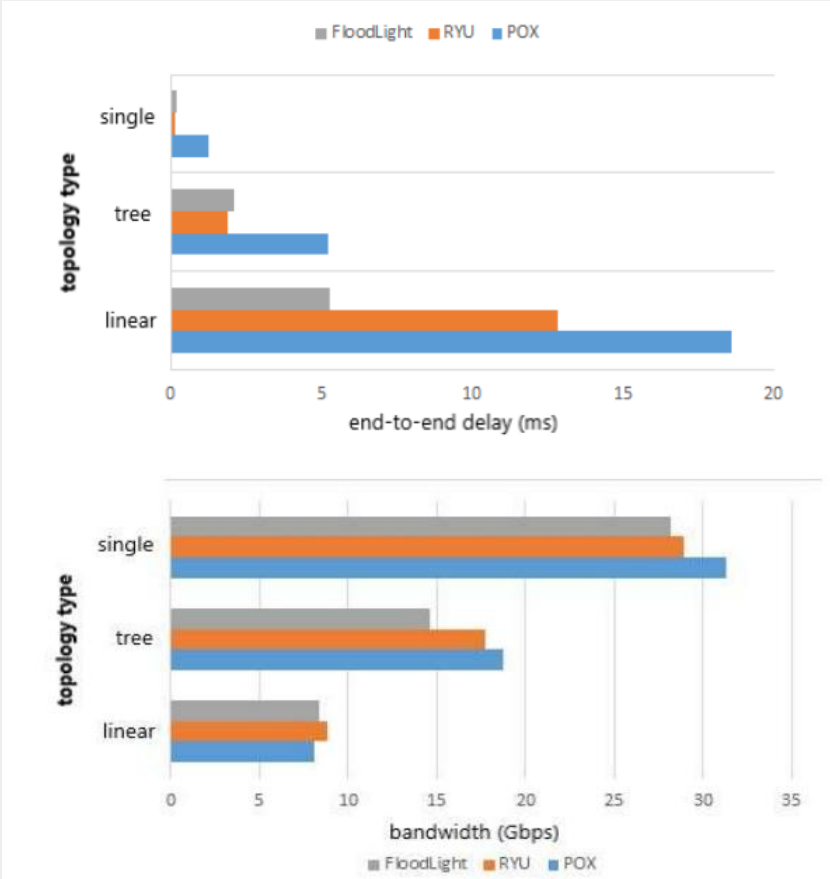


Effective Network Variables on Performance

MTU

Data Center Needs





Fat Tree Topology

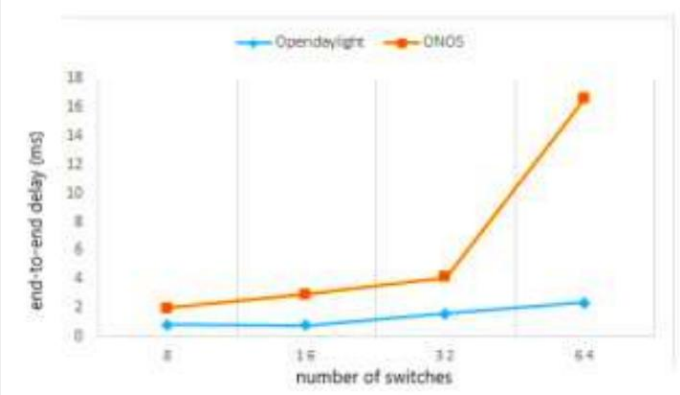
Why We Need Fat Tree Topology

Google Fat Tree & Face Book Fat Tree

Fault Tolerability

ONOS & OpenDayLight

Winner For Fat Tree Topology



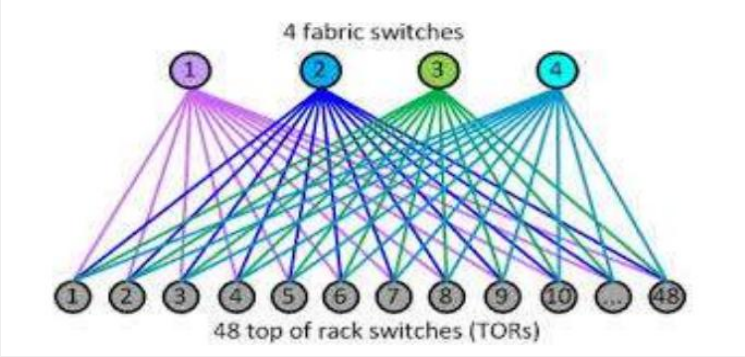


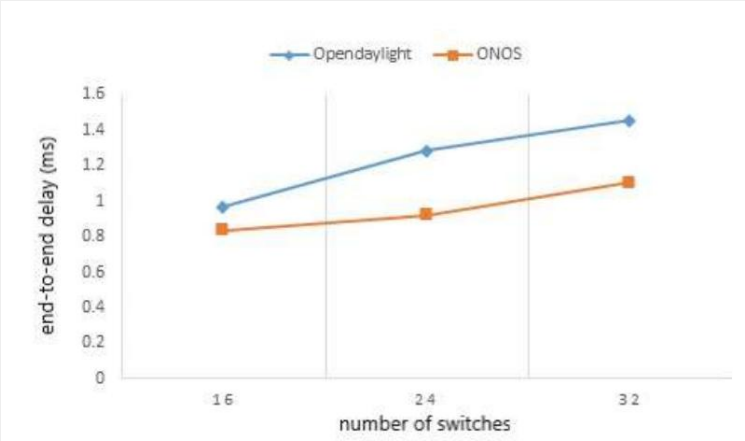
Facebook Fat Tree Topology

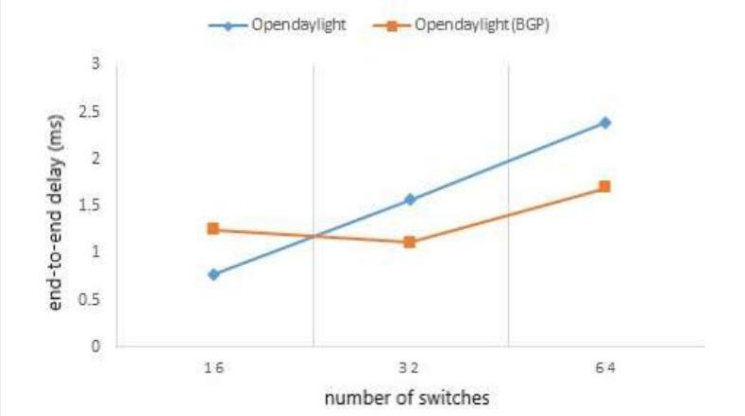
ONOS and OpenDayLight

TOR

Who is the Winner Overall







Conclusion

We Studied

- Summarization Of How SDN Work
- Comprehensive Study Of Open Flow And Controller Centralized Or Not
- Categorizing Two Centralized Controllers (Small networks) Distributed Controllers (Data Centers)

Parameters Of Study

- QoS → Latency , Throughput , Topologies , Scales
- End To End Delay With Different Topologies Like Google Fat Tree
- Different Work Loads

What We Found Out

- Best in Network Management in large scales & Centralized in terms of QoS
- Best in Network Management in small Scales & Centralized in term of QoS
- Best in Tree Topology
- Best in Linear Topology
- Best Centralized In Term Of Least Delay In Large Scale Network
- Highest Trough-put than Others
- Best Distributed In Google Fat tree
- Best Distributed In Facebook fat tree
- Stable Distributed In Network With Increasing Work Load Without Much Increased Delay

با تشکر!

Sources :

Article Source :

Performance Evaluation Of Software Defined Networks A Comparative Study

https://www.sid.ir/paper/975866/en

GAPGPT AI

https://gapgpt.app/chat

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